

Section 2

2025年12月9日 星期二

14:03

DECA S2

5th December, 2025.

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Reference: DECA Part 2, section 2.

Aim: Constructing logics of a vending machine, then add LED lights to show status; and add an input V_c to indicate whether the machine is in progress for challenge.

Procedure

• Before this lab.

The vending machine has a transition diagram like this, where we have 3 inputs: C ; S ; Condition ($\beta_1\beta_0$, 2-bits).

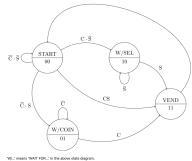


Fig2.1

I need to generate boolean expressions, for NEXT condition ($\beta_1'\beta_0'$) for the machine.

• Draw a truth table.

β_1	β_0	C	S	$\beta_1'\beta_0'$
0	0	1	0	0
0	0	0	1	0
0	0	0	0	0
0	0	1	1	1
0	1	0	X	0
0	1	1	X	1
1	0	X	0	1
1	0	X	1	1
1	1	X	X	0

Fig2.2

• Draw a Karnaugh map for β_1' and β_0' to get expressions.

β_1'	β_0'	C	S
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

Fig2.3

$$\beta_1' = \beta_1\beta_0 + \beta_1C$$

β_0'	β_1'	C	S
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

Fig2.4

$$\beta_0' = \beta_1\beta_0 + \beta_0S$$

$$\beta_1' = \beta_1\beta_0 + \beta_1C$$

$$\beta_0' = \beta_1\beta_0 + \beta_0S$$

Logics are like:

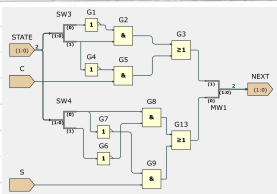


Fig2.5

• State machine.

Connecting NEXT to a register, which stores data, a simple vending machine is made.

Testing: I expect following case:

00 → select ($\beta_1\beta_0$) 01 → β_1 in (β_1) 11 → Restart 00.

and testing is same:

hex dec bin

Clock Tick 1

Inputs

S

C

Outputs & Viewers

OUT (2 bits)

Stateful components

Register: REG1 (2 bits)

hex dec bin

Clock Tick 2

Inputs

S

C

Outputs & Viewers

OUT (2 bits)

Stateful components

Register: REG1 (2 bits)

hex dec bin

Clock Tick 3

Inputs

S

C

Outputs & Viewers

OUT (2 bits)

Stateful components

Register: REG1 (2 bits)

hex dec bin

Clock Tick 4

Inputs

S

C

Outputs & Viewers

OUT (2 bits)

Stateful components

Register: REG1 (2 bits)

• Outputs.

We now need to add 2 things.

①. A motor, when coins and select are done, we should activate a motor to vend the product, then the machine restarts.

②. An LEDC and an LEDS. When S isn't done, LEDS should be on to remind customers, after S is done, LEDS is off. (same for C & LEDC).

which is as:

State	M	LEDS	LEDC
START	0	1	1
W/SEL	0	1	0
W/COIN	0	0	1
VEND.	1	0	0

The inputs now are $\bar{A}_1, \bar{A}_0$, again we draw truth table.

$\bar{A}_1$	$\bar{A}_0$	M	LEDS	LEDC
0	0	0	1	1
0	1	0	1	0
1	0	0	0	1
1	1	1	0	0

$\therefore M = \bar{A}_1 \bar{A}_0$

$$LEDS = \bar{A}_1 \bar{A}_0 + \bar{A}_1 \bar{A}_0 = \bar{A}_1 (\bar{A}_0 + \bar{A}_0) = \bar{A}_1$$

$$LEDC = \bar{A}_1 \bar{A}_0 + \bar{A}_1 \bar{A}_0 = \bar{A}_0 (\bar{A}_1 + \bar{A}_1) = \bar{A}_0$$

$\therefore$ logics are like:

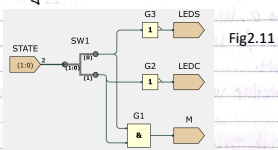


Fig2.11

Add to former vending machine.

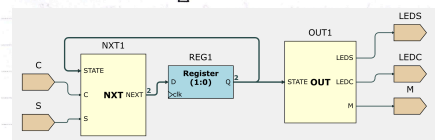


Fig2.12

Testing, I expect following:

LEDS: 1, LEDC: 1, M: 0 $\rightarrow$ LEDS: 0, LEDC: 1, M: 0 $\rightarrow$ LEDS: 0, LEDC: 0, M: 1 $\rightarrow$ LEDS: 1, LEDC: 0, M: 0

Testing is the same.



• Challenge: Extending the vending machine:

Now we need to add an input V . When the vending machine is in progress, V will be 0. When the machine reaches VEND, which is 11, it will wait for V , only if V reaches 1, the machine will then restart.

V is for monitoring whether vending is complete and controlling whether should restart.

①. Again draw a truth table.

there're $\bar{A}_1, \bar{A}_0, C, S, V \rightarrow 5$ inputs now.

$\bar{A}_1$	$\bar{A}_0$	C	S	V	$\bar{A}_1$	$\bar{A}_0$
0	0	0	0	x	0	0
0	0	0	1	x	0	1
0	0	1	0	x	1	0
0	0	1	1	x	1	1
0	1	0	x	x	0	1
0	1	1	x	x	1	1
1	0	x	0	x	1	0
1	0	x	1	x	1	1
1	1	x	x	0	1	1
1	1	x	x	1	0	0

Fig2.17

As you can see, the first 6 rows brings no differences as V is just a non-relevant term.

So we just add results from first 6 terms to 7th & 8th term.

$$\bar{A}_1 = \bar{A}_1 \bar{A}_0 + \bar{A}_1 C + \bar{A}_1 \bar{A}_0 V; \bar{A}_0 = \bar{A}_1 \bar{A}_0 + \bar{A}_0 S + \bar{A}_1 \bar{A}_0 V$$

$$= \bar{A}_1 (\bar{A}_0 + \bar{A}_0 V) + \bar{A}_1 C = \bar{A}_0 (\bar{A}_1 + \bar{A}_1 V) + \bar{A}_0 S$$

$$= \bar{A}_1 (\bar{A}_0 + \bar{A}_0 V) + \bar{A}_1 C = \bar{A}_0 (\bar{A}_1 + \bar{A}_1 V) + \bar{A}_0 S$$

So the logics are.



Fig2.18

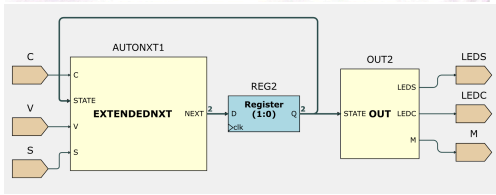
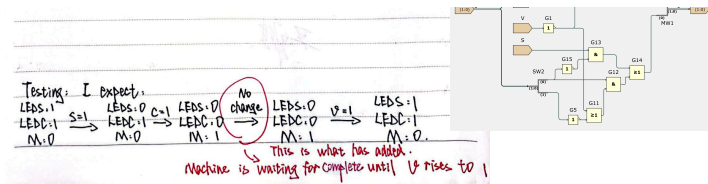


Fig2.19

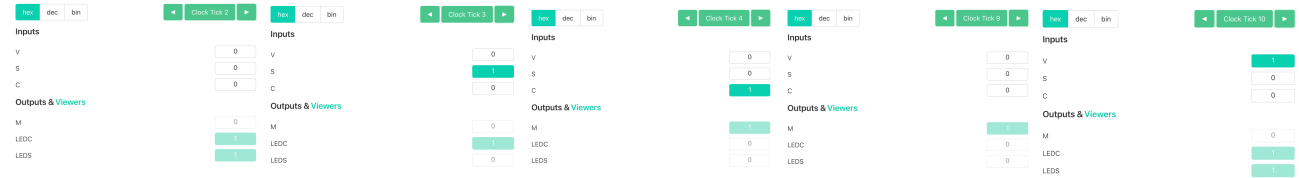


Fig2.20~2.24

Testing is as expected

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